MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 7381

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Dual TR (bandpass) used with short-slot hybrid couplers

ABSOLUTE RATINGS:

Parameter:	Transmitter po	Frequency	Ignitor voltage	Operating temperature
Unit:	kw	MHz	Vdc	°C
Minimum:	10-3	8,490	-800	-55
Maximum:	200	9,610	---	125

PHYSICAL CHARACTERISTICS: See figure 1

TEST CONDITIONS:

Parameter:	Transmitter po	Ii	tp	prr	Ri
Unit:	kw	μAdc	μs	pps	Meg
Tolerance	±10%	±3%	±1.5%	±5%	5%
	40	100	1.0	1000	4.9

TEST FREQUENCIES	
F	MHz ±0.1%
F_1	8490
F_2	8600
F_3	8800
F_4	9000
F_5	9200
F_6	9400
F_7	9610

REQUIREMENT OR TEST:

GENERAL

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection					
1002	Barometric pressure, reduced	Ebb = -800 Vdc; R1 = 4.9 Meg each ignitor; pressure = 10.1 psia (max)	---	---	---	---
4021	Degradation due to vibration	t = 15 minutes	---	---	---	---
4491	Transmitter-Receiver isolation	F = F1 and F7 F = F2, F3, F4, F5 and F6 I1 = 100 μAdc on each ignitor	Ratio Ratio	10 15	--- ---	db db
4474	High-level VSWR	po = 20 kw ±10%; F = F4 ±5%	---	---	1.2	---
4452	Spike leakage energy (2)	T = TA (see note 4) T = 125°C (see note 4) T = 125°C (see note 5) tp_1 = 1.0 ±0.15 μs; tp_2 = 0.5 ±0.15 μs; F = F1 and F7	Ws Ws Ws	--- --- ---	0.3 0.5 0.15	ergs ergs ergs
4452	Flat leakage power (2)	T = TA (see note 4) T = 125°C (see note 4) T = 125°C (see note 5) tp_1 = 1.0 ±0.15 μs; tp_2 = 0.5 ±0.15 μs; F = F1 and F7	po po po	--- --- ---	80 150 15	mW mW mW
	Quality conformance inspection, part 1					
4006	Temperature cycling	-55°C to +125°C (see note 1)	---	---	---	---
4401	Ignitor ignition time	Ebb = -800 Vdc; R1 = 4.9 Meg each ignitor	t	---	5	sec
4406	Ignitor voltage drop	I1 = 100 μAdc on each ignitor	Eid	200	450	Vdc
4473	Low-level VSWR	F = F1, F4 and F7	---	---	1.2	---
4489	Duplexer loss	F = F4	Li	---	1.1	db
4452	Spike leakage energy (1)	F = F4 T = TA (see note 4) T = 125°C (see note 5) tp_1 = 1.0 ±0.15 μs; tp_2 = 0.5 ±0.15 μs	Ws Ws	--- ---	0.3 0.15	ergs ergs
4452	Flat leakage power (1)	F = F4 T = TA (see note 4) T = 125°C (see note 5) tp_1 = 1.0 ±0.15 μs; tp_2 = 0.5 ±0.15 μs	po po	--- ---	80 15	mW mW

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2					
----	Pressure operation	50 psia (min) (see note 3)	---	---	---	--
4460	Ignitor noise ratio	F = F1 +30 MHz; F = F4; F = F7 -30 MHz; I1 = 100 μAdc (min)	Nr	---	1.3	--
4486	Arc loss	F = F4	L1	---	0.8	db
4021	Vibration (2)	F = 45 to 55 Hz; 20 G; t = 10 minutes; I1 = 100 μAdc on each ignitor (see note 8)	Eid	100	---	Vdc
4473	Low-level VSWR	F = F2, F3, F4, and F5	---	---	1.2	--
4489	Duplexer loss	F = F1 and F7	L1	---	1.2	db
4452	Spike leakage energy (2)	T = 125°C (see note 4); I1 = 100 μAdc on each ignitor	Ws	---	0.5	ergs
4452	Flat leakage power (2)	T = 125°C (see note 4); I1 = 100 μAdc on each ignitor	po	---	150	mW
4471	Recovery time	T = TA and 125°C; po = 150 kw ±10%; tp = 1.0 μs ±0.05 μs; F = F4 ±5%	t	---	3.0	μs
	Quality conformance inspection, part 3					
4006	Temperature-cycling life test	T = -55°C to +125°C; Ebb = -800 Vdc; R1 = 4.9 Meg each ignitor; Group C; 10 Hz (min) (see note 2)	L1	50	---	μAdc
----	Life-test provisions	po = 150 kw ±5% Eff = -800 Vdc; Ri = 4.9 Meg on each ignitor; t = 100 hours at 125°C and 400 hours at 85°C; Group D (see note 6)	---	---	---	--
----	Life-test end points:					
4406	Ignitor voltage drop (1)	I1 = 100 μAdc	Eid	225	475	Vdc
4471	Recovery time	T = TA po = 150 kw tp = 1.0 μs ±.05%	t	---	8	μs
4489	Duplexer	F = F4	L1	---	1.2	db
4452	Spike leakage energy	T = 125°C (see note 4)	Ws	---	0.5	erg
----	Mixer NF degradation	(see note 6)	NF	---	2.0	db

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NOTES:

1. The tube shall be subjected to the number of temperature cycles and temperature limits specified. The tube shall be maintained at each end-point temperature for sufficient time to reach equilibrium, but not for less than 5 minutes. Changes in temperature from end point to end point shall be gradual but the cycle shall not exceed 30 minutes. After cycling, the tube shall be allowed to return to room temperature, held for 24 hours, and then shall be tested for the specified limits of ignitor ignition time and ignitor voltage drop.

2. The minimum temperature rate of change shall be 2.5°C/minute. The test shall be conducted with aluminum flanges (see figure 2) securely attached to each end of the tube. The ignitor current shall be monitored during the test and shall remain within the limit specified.

3. The air pressure specified shall be applied to the input and output window of the tube for a period of 2 hours, after which the pressure shall be reduced to atmospheric pressure. The pressure is permitted to decrease by 10 percent of the specified value during the test interval. After this test, the tube shall pass the ignitor voltage test.

4. The leakage energy shall be measured for each channel individually.

5. The leakage energy shall be measured for the receiver arm as shown in Test Circuit a.

6. During the life test, each channel shall be terminated with JAN 1N23WE diode mounted in a balanced mixer. The diode will satisfy MIL-S-19500/233 of the latest issue. The mixers will be mounted as shown in Test Circuit b. The local oscillator input during life shall be terminated with 100-ohm video loads. When tubes are being life tested at elevated temperatures, the diode shall be maintained at 25°C nominal.

   At the beginning of the life test, the balanced mixers shall each have a noise figure less than 8.0 db. The mixer noise figures shall be remeasured periodically during the course of the test. The interval between measurement shall be not more than 100 hours. Should diode failure occur, the failed diode shall be replaced and the life test continued. If a diode failure occurs during the last 100 hours of the required life test, the failed diode shall be replaced and the life test continued for a minimum of 100 additional hours. However, only one diode failure per each balanced mixer shall be allowed during the life test. If a failure of the test equipment (keep-alive-power supply, etc.) occurs and damages all diodes, the diodes shall be replaced and the life test continued. If the test equipment failure occurs during the last 100 hours of required life test, the diodes shall be replaced and the life test continued for a minimum of 100 additional hours. The noise figure of the two balanced mixers must remain within the specified limits throughout the life test.

7. All measurements that include short-slot hybrids shall be made with gaskets bolted between input and output flanges of tube and mating flanges.

8. The tube shall be vibrated in a plane parallel to the ignitor axis under the conditions specified. Input flange shall be rigidly coupled to the vibrating table, either coupled directly for horizontally vibrating table, or coupled through an "L" shaped mount for vertically vibrating table. The tube shall be positioned so that both ignitors are equidistant from the vibrating table surface. The output flange shall be rigidly connected to a standard 12 ± 1/2-inch dual length of RG52U bronze waveguide with standard dual flanges at each end. Vibration frequency shall be adjusted in the range specified until resonance is obtained. Vibration shall continue at the resonant frequency for the specified time. The ignitor voltage drop shall be monitored during the year.

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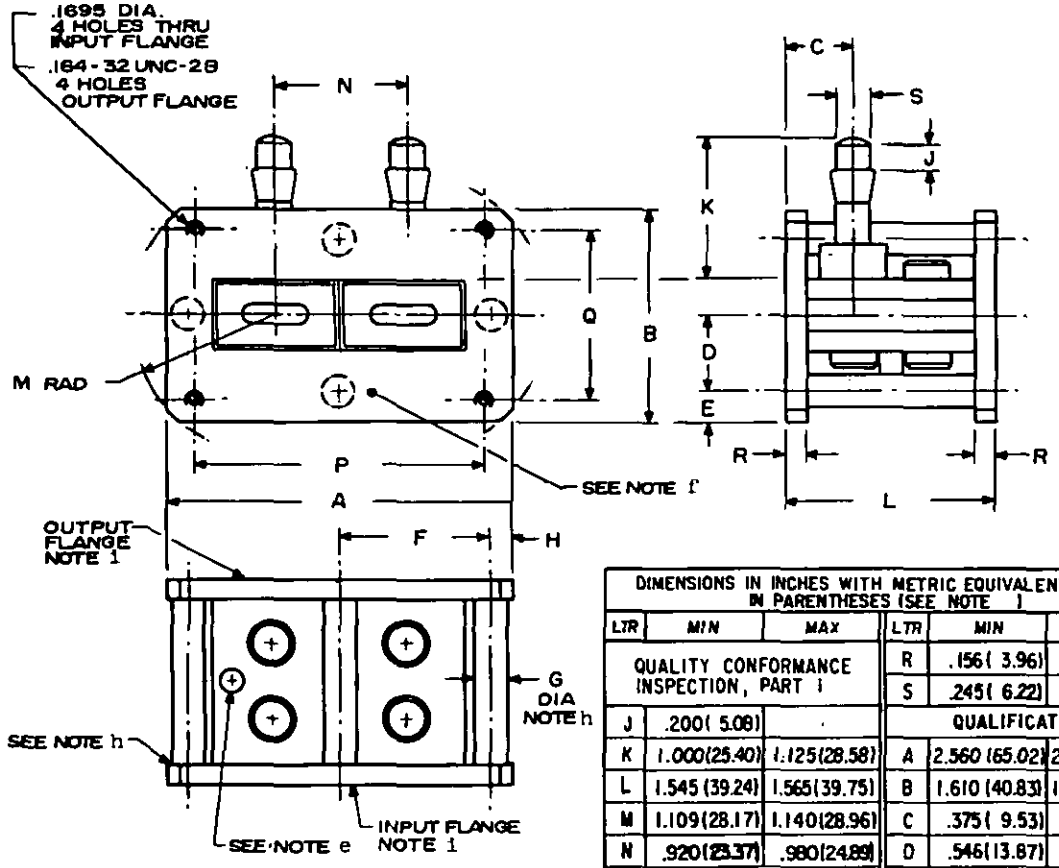
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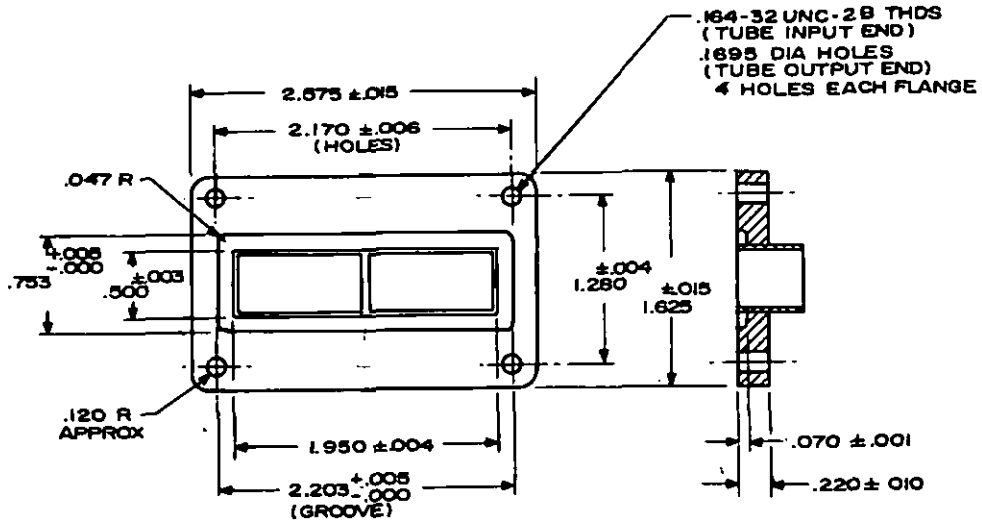
DIMENSIONS IN INCHES WITH METRIC EQUIVALENTS (mm) IN PARENTHESES (SEE NOTE)					
LTR	MIN	MAX	LTR	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART 1			R	.156 (3.96)	
			S	.245 (6.22)	.255 (6.48)
J	.200 (5.08)		QUALIFICATION		
K	1.000 (25.40)	1.125 (28.58)	A	2.560 (65.02)	2.590 (65.79)
L	1.545 (39.24)	1.565 (39.75)	B	1.610 (40.83)	1.640 (41.66)
M	1.109 (28.17)	1.140 (28.96)	C	.375 (9.53)	
N	.920 (23.37)	.980 (24.89)	D	.546 (13.87)	.578 (14.68)
QUALITY CONFORMANCE INSPECTION, PART 2			E	.234 (5.94)	.265 (6.73)
			F	1.109 (28.17)	1.140 (28.96)
P	2.165 (54.99)	2.175 (55.25)	G	.234 (5.94)	.265 (6.73)
Q	1.276 (32.40)	1.284 (32.61)	H	.140 (3.56)	.171 (4.34)

NOTES:

a. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
b. The rectangle formed by the four mounting holes on each flange shall be centered on the flange face within .031 (.79 mm).
c. Silver plate 100 MSI, or equivalent.
d. Rhodium flash over silver plate optional.
e. Exhaust tube shall not extend beyond flange and shall not block mounting holes.
f. Input flange of tube shall be flat within .004 (.10 mm).
g. Finish: black paladin primer followed by optical black baking enamel, or approved equivalent, to all outside surfaces except contacting surfaces of flanges, holes and threads of flanges and electrode insulators and caps.
h. Four .250 (6.35 mm) diameter rods brazed or welded to flanges in eight places on tube flanges spaced as shown on figure 1. Rods for mechanical support only.
i. Mark "input" or "output" on edge of each flange as appropriate.

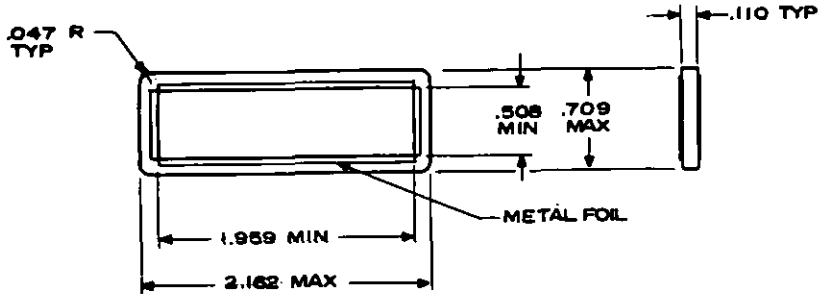
FIGURE 1. Outline drawing of electron tube type 7381.

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MATING FLANGE

1- FLANGE FOR INPUT END
1- FLANGE FOR OUTPUT END



GASKET ASSEMBLY

2- REQUIRED PER TUBE

INCHES	MM	INCHES	MM
.001	.03	.220	5.59
.003	.08	.500	12.70
.004	.10	.508	12.90
.005	.13	.709	18.01
.006	.15	.753	19.13
.010	.25	1.280	32.51
.015	.38	1.625	41.28
.047	1.19	1.950	49.53
.070	1.78	1.959	49.76
.110	2.79	2.162	54.91
.120	3.05	2.170	55.12
.1695	4.31	2.575	65.41

FIGURE 2 Mating flange and gasket assembly

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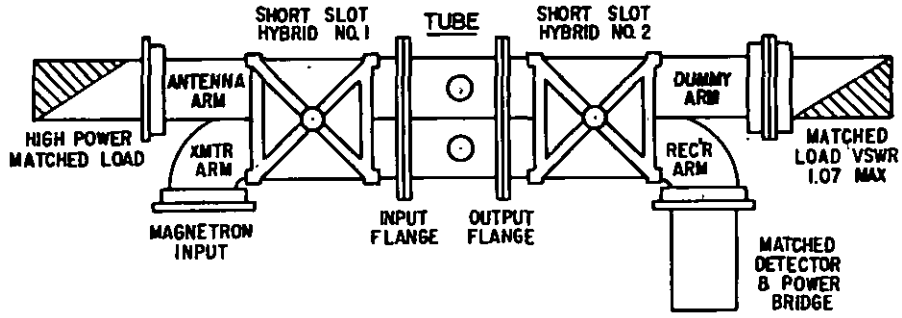


FIGURE 3 Test Circuit a

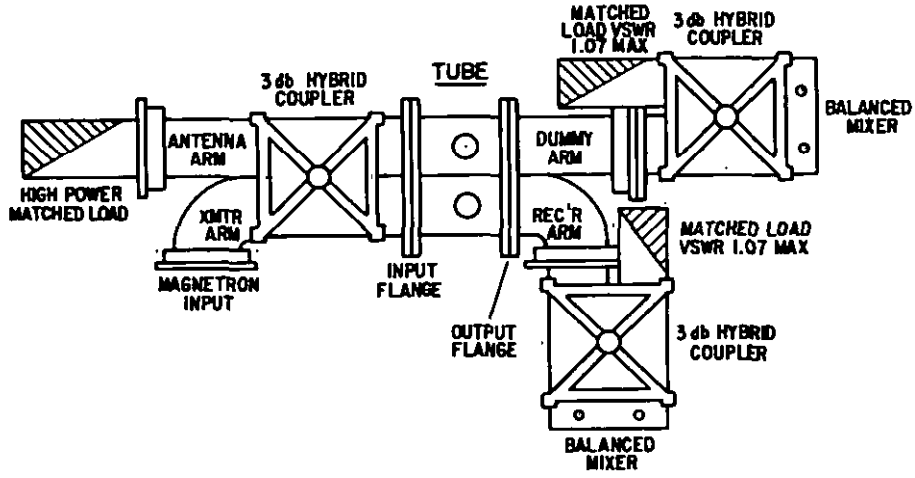


FIGURE 4 Test Circuit b

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